

RadFinder – Learning to Read Where to Look: Disease-Aware Vision-Language Pretraining for 3D CT

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Radiology VLMs

- Current models use contrastive learning for scan-report alignment

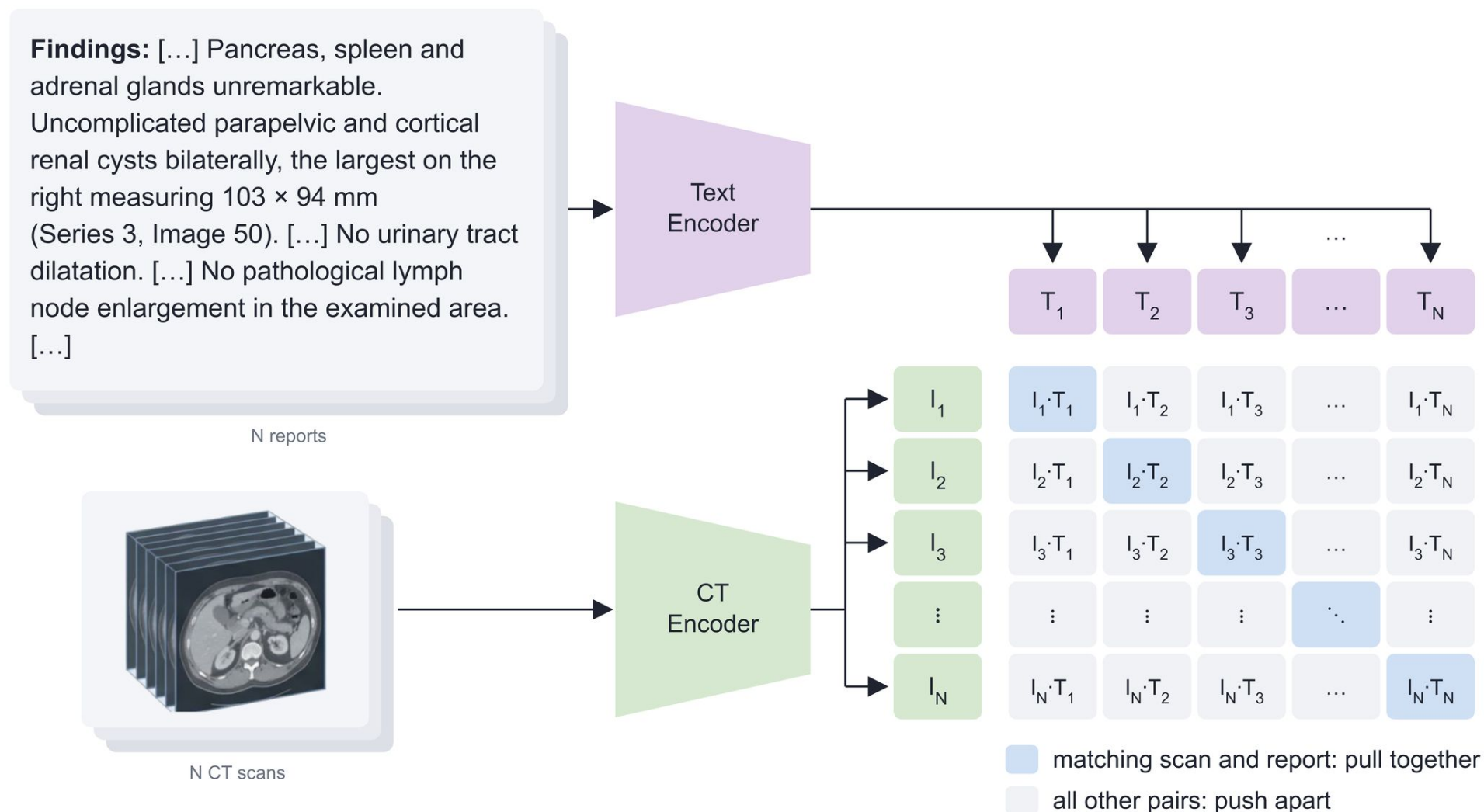


Figure based on:

Radford et al., Learning Transferable Visual Models From Natural Language Supervision, ICML 2021

Claessens et al., Scaling Self-Supervised and Cross-Modal Pretraining for Volumetric CT Transformers, arXiv:2511.17209, 2025

What supervision do radiology reports contain?

Findings: [...] Pancreas, spleen and adrenal glands unremarkable.
Uncomplicated parapelvic and cortical renal cysts bilaterally, the largest on the right measuring 103 × 94 mm. [...]
No urinary tract dilatation. [...] No pathological lymph node enlargement in the examined area. [...]

report

- Whole report describes the scan

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*Simple renal cysts: **Yes***

*Hydronephrosis: **No***

report

- Whole report describes the scan
- Sentences state findings

Radiologists tell us where to look

Findings: [...] Pancreas, spleen and adrenal glands unremarkable.

Uncomplicated parapelvic and cortical renal cysts bilaterally, the largest on the right measuring 103 × 94 mm (**Series 3, Image 50**). [...]

No urinary tract dilatation. [...] No pathological lymph node enlargement in the examined area. [...]

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Snippet

report

- Whole report describes the scan
- Sentences state findings
- Snippets contain image references

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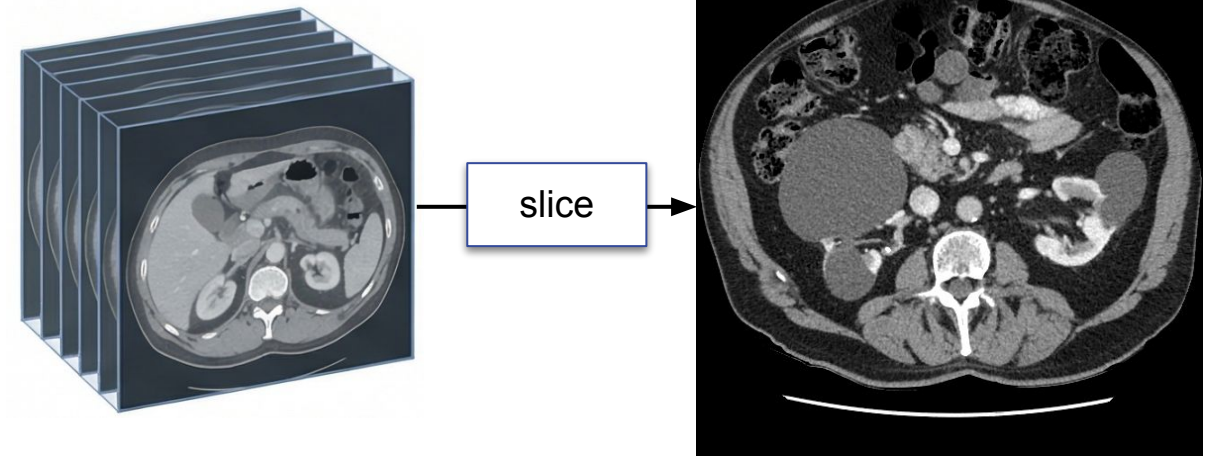
Simple renal cysts: Yes

Hydronephrosis: No

Snippet

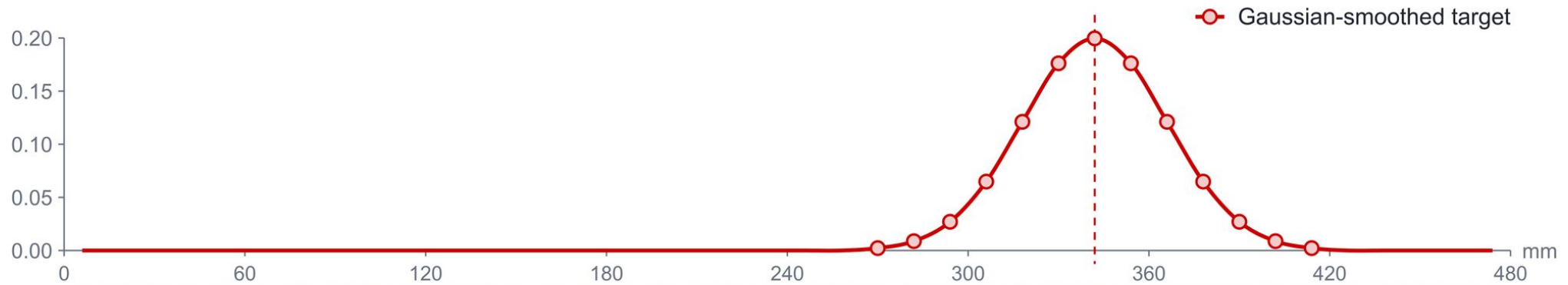
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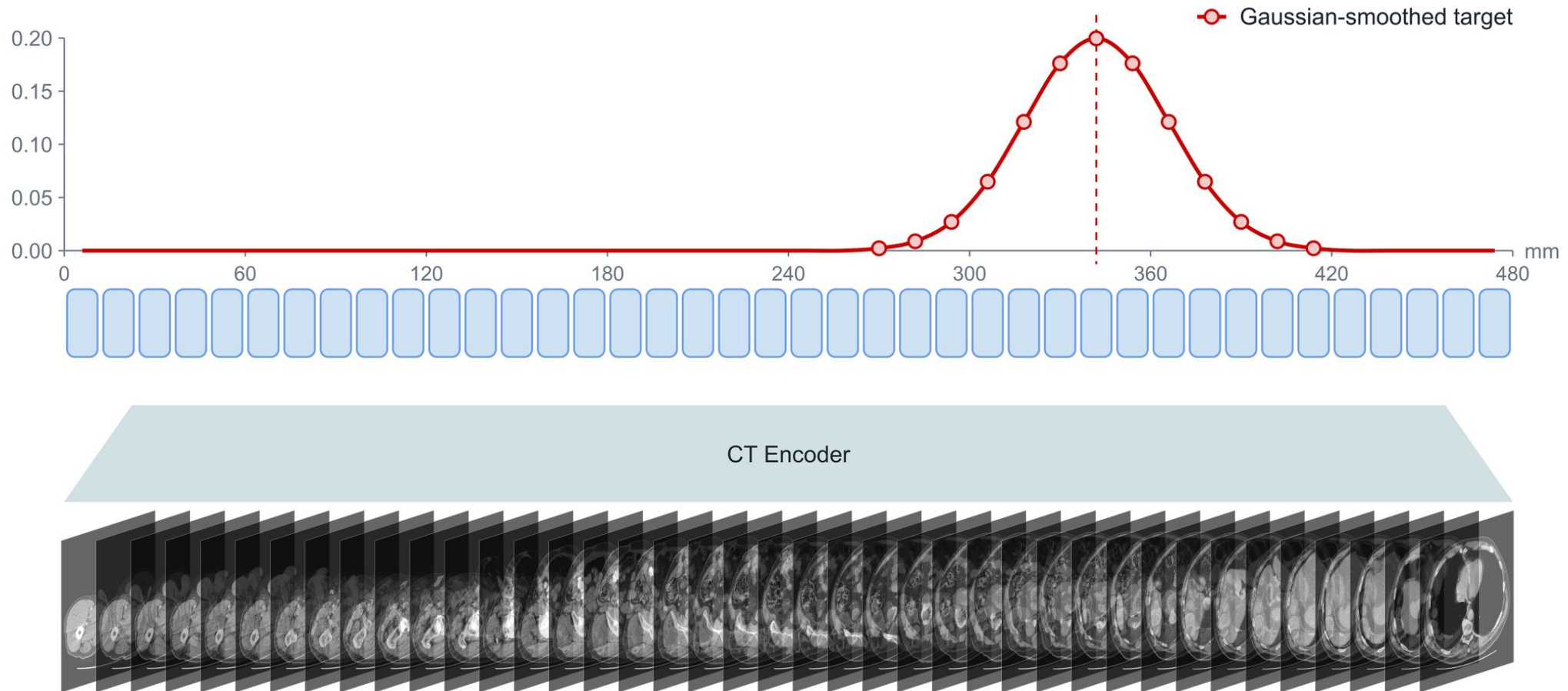
Learning to localize

Uncomplicated parapelvic and cortical renal cysts bilaterally, the largest on the right measuring 103 × 94 mm



Learning to localize

Uncomplicated parapelvic and cortical renal cysts bilaterally, the largest on the right measuring 103 × 94 mm

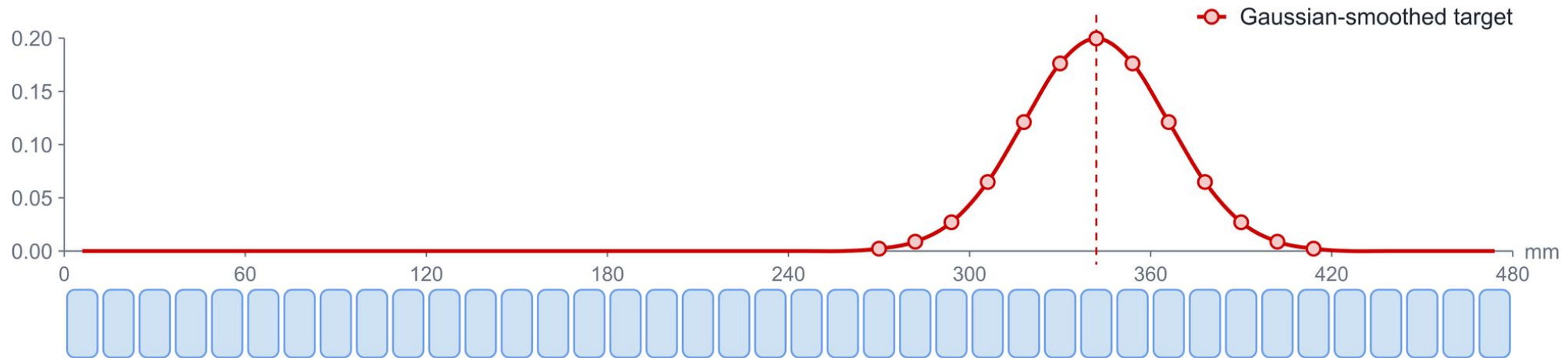


Learning to localize

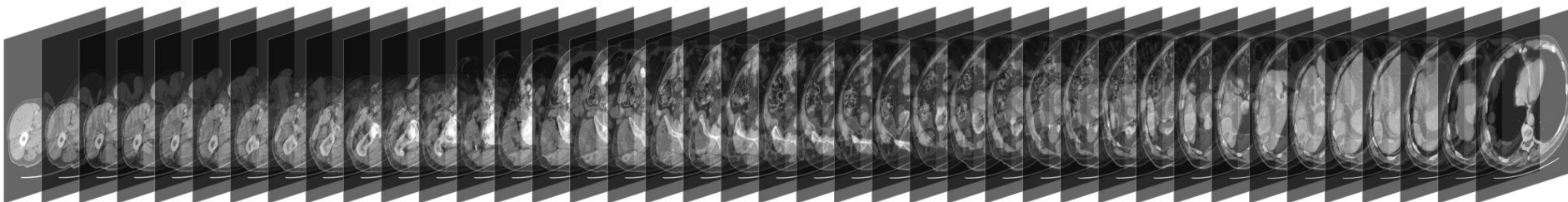
Uncomplicated parapelvic and cortical renal cysts bilaterally,
the largest on the right measuring 103 × 94 mm

Text
Encoder

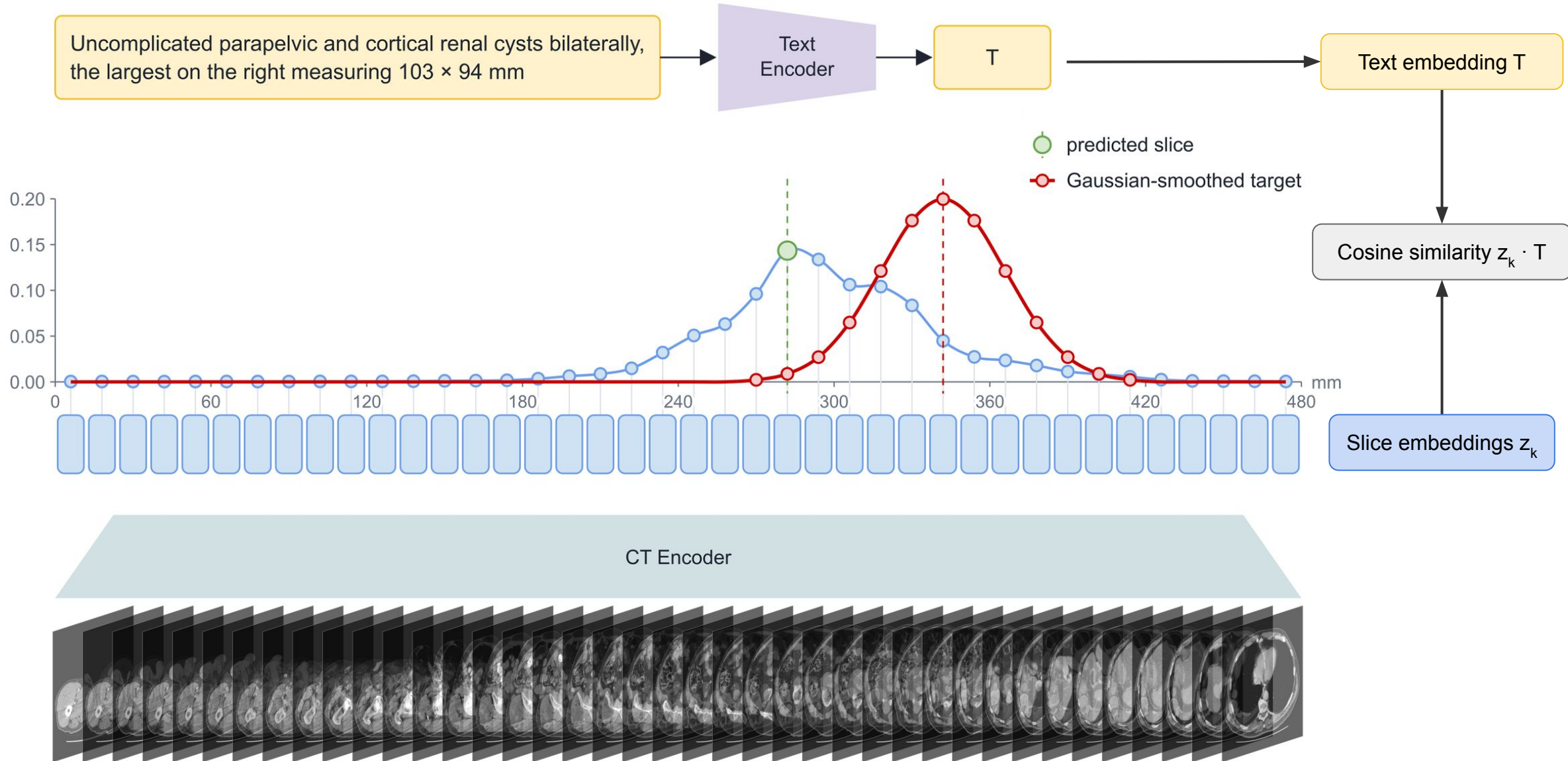
T



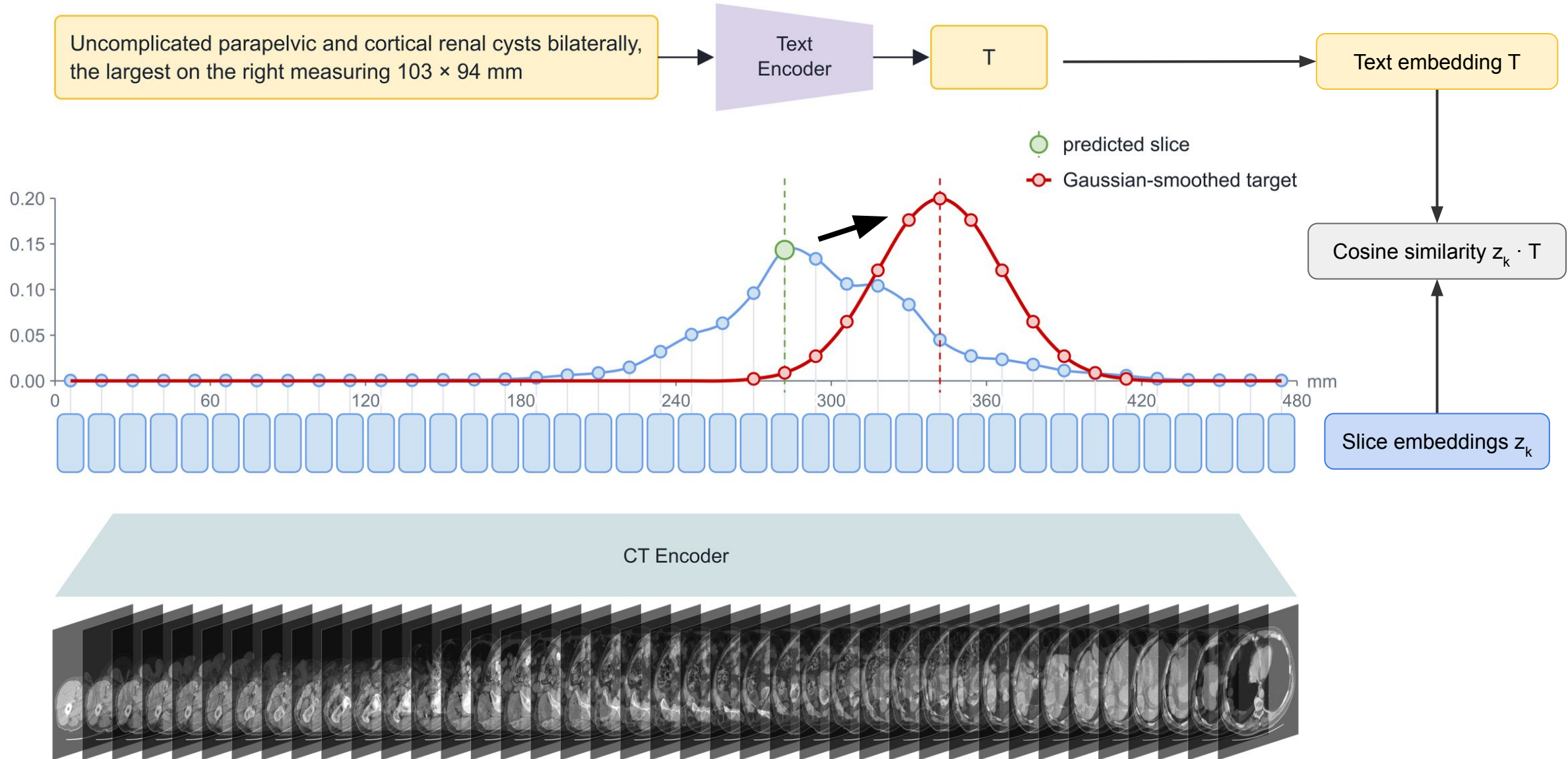
CT Encoder



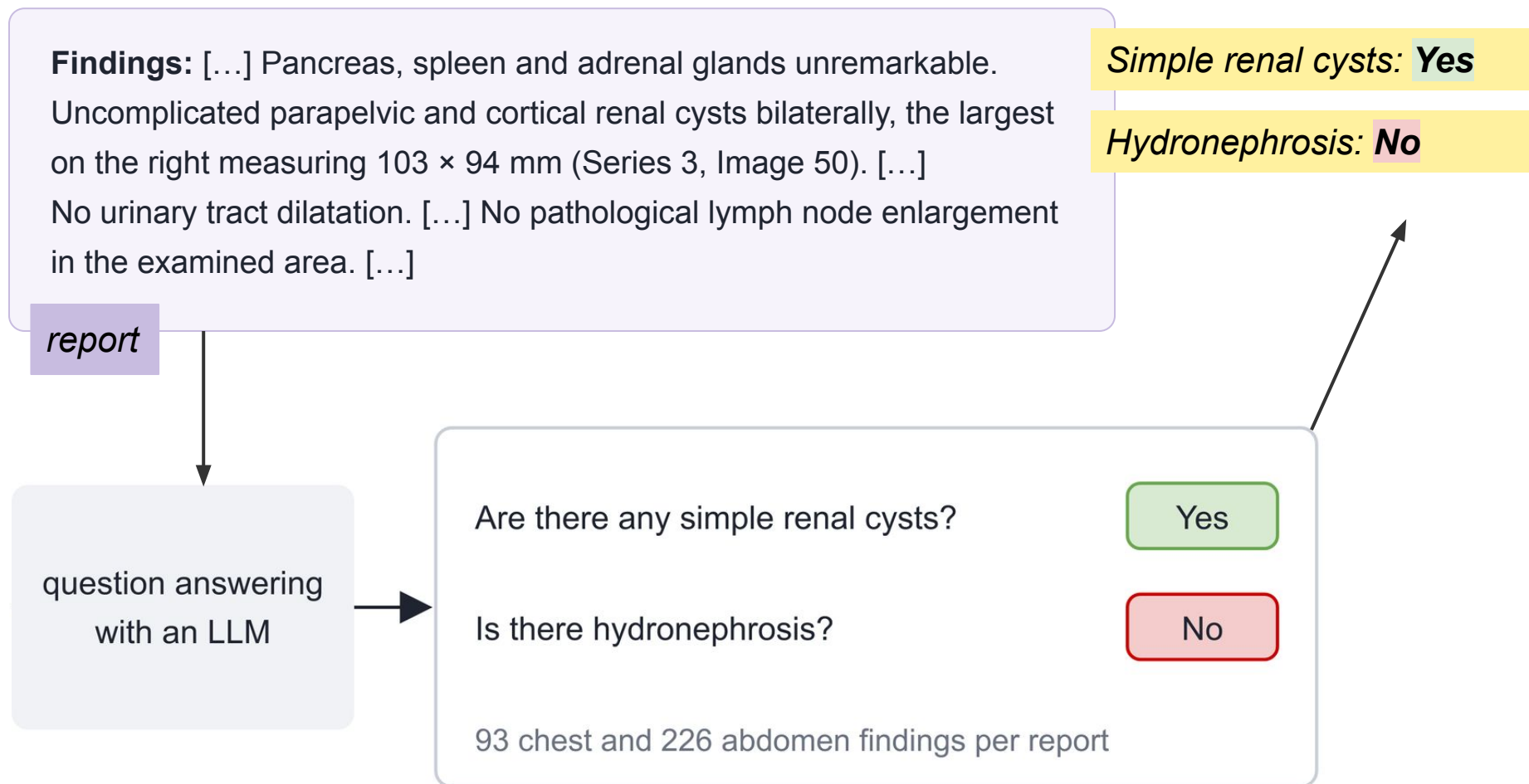
Learning to localize



Learning to localize



Disease label extraction



Radiology Text Engine (RATE) pipeline (Pillar-0, Agrawal et al., 2025)

Prompt-based disease label training

Simple renal cysts

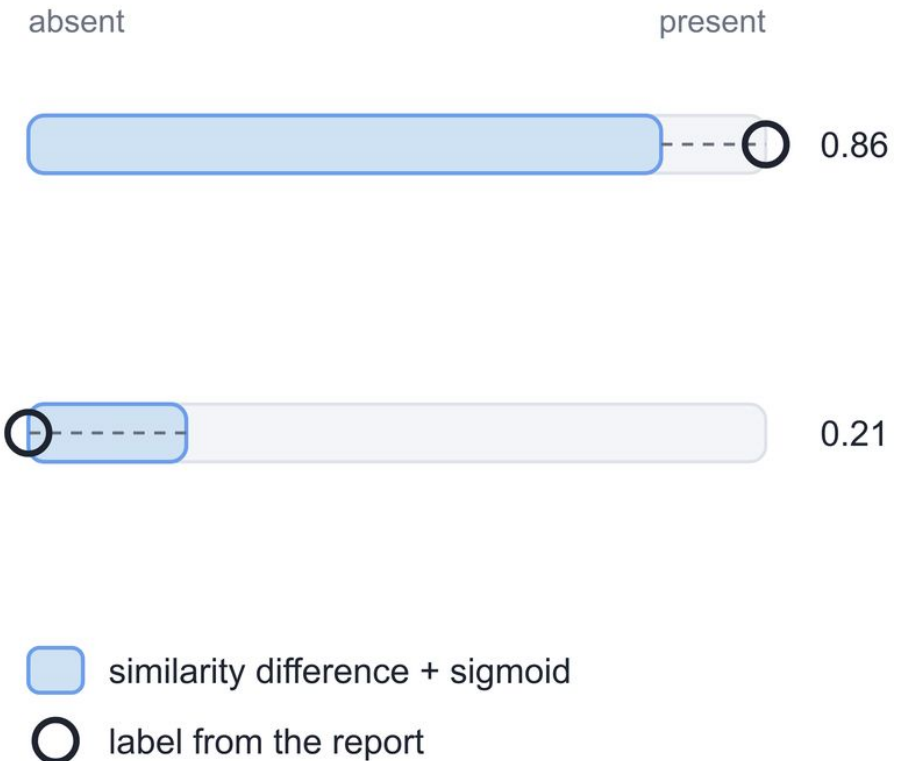
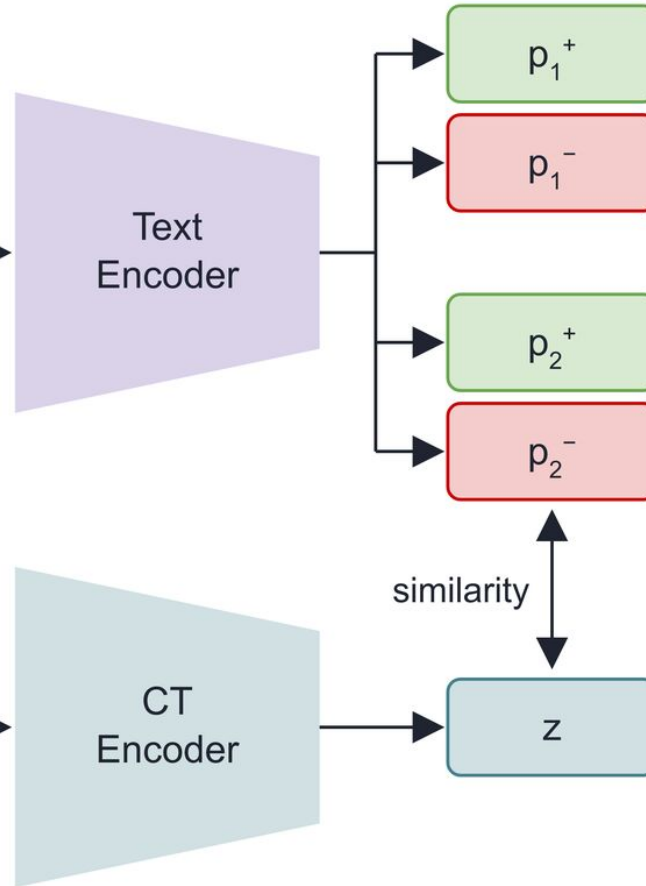
Simple renal cysts are present.

There are no simple renal cysts.

Hydronephrosis

Hydronephrosis is present.

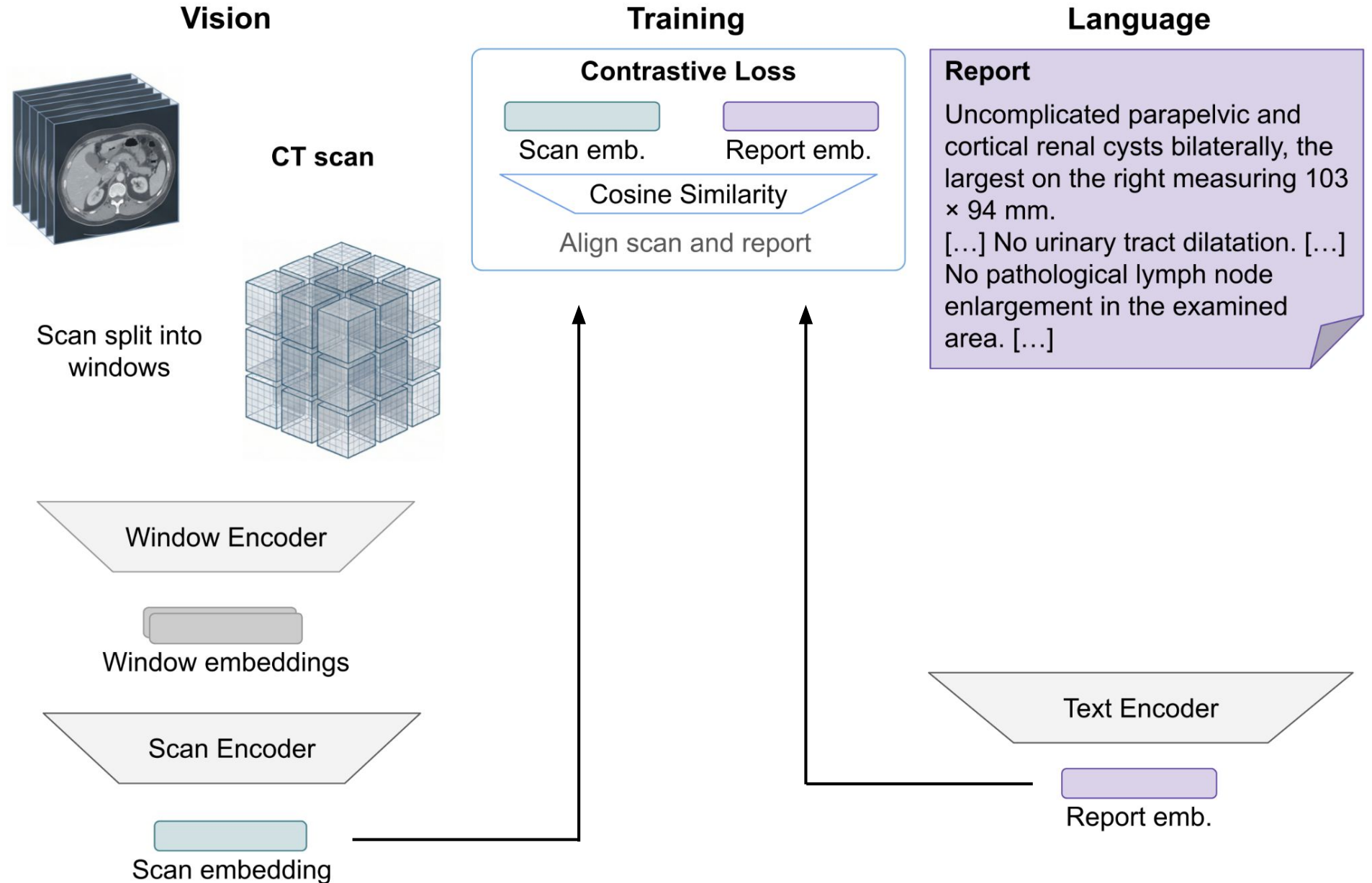
There is no hydronephrosis.



Using label prompts in contrastive training (UniCL, Yang et al. CVPR 2022 & CXR-CLIP, You et al., MICCAI 2023)

Base model

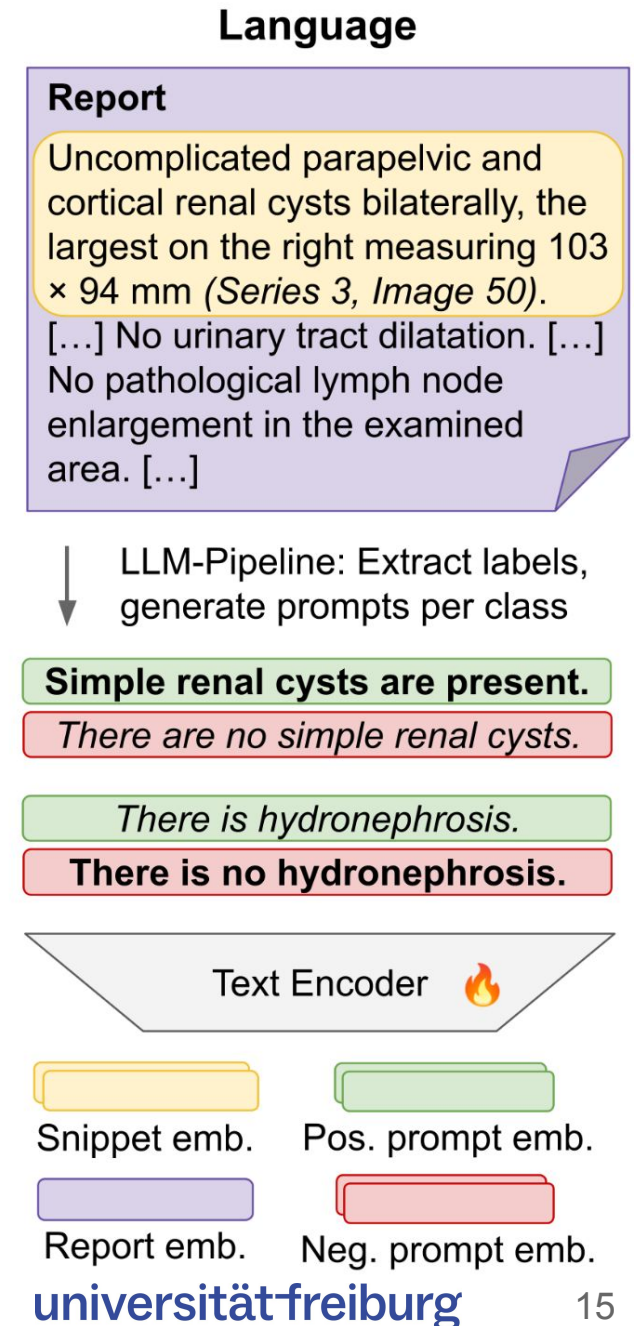
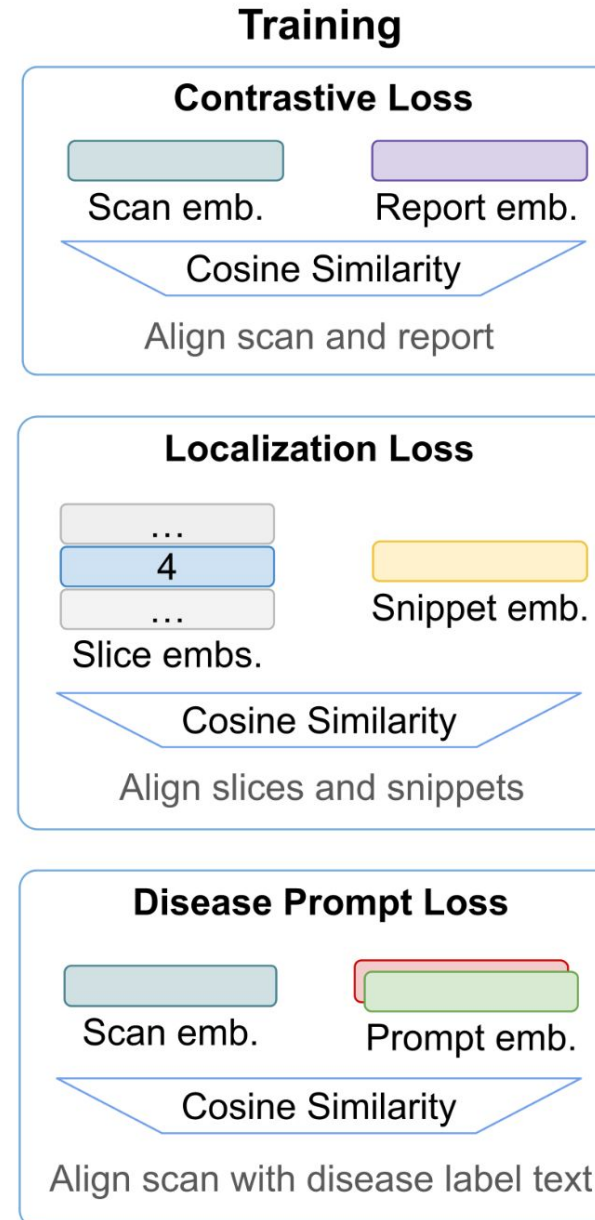
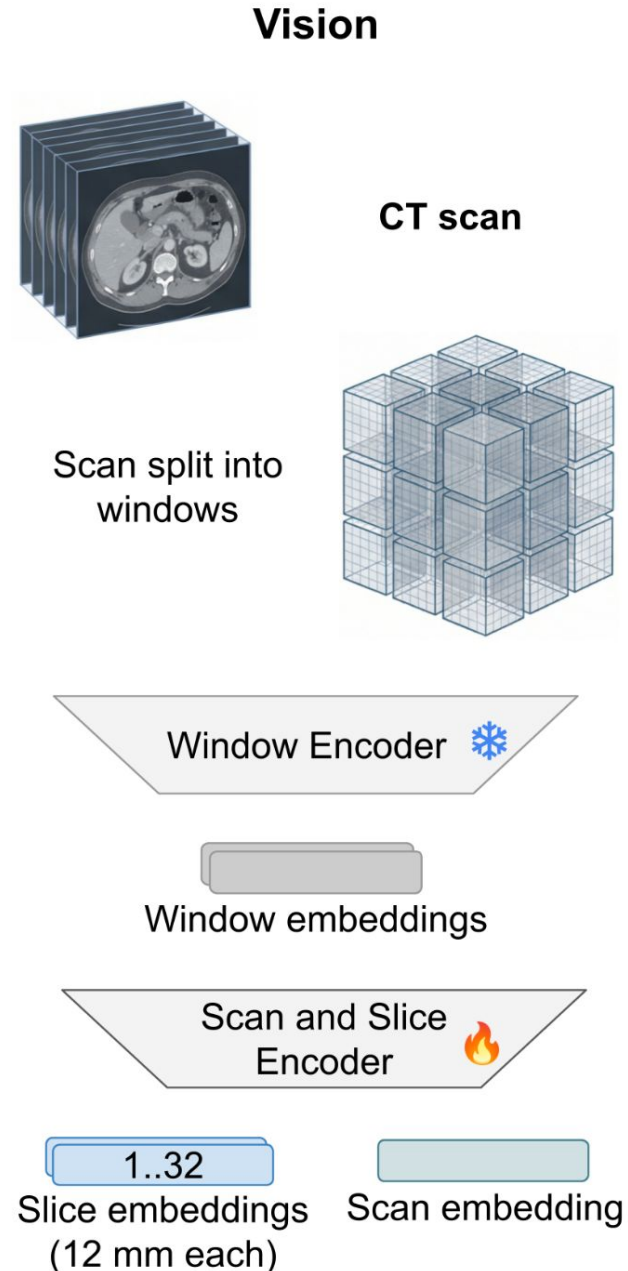
SPECTRE (Claessens et al., CVPR 2026)



RadFinder

SPECTRE fine-tuned with

- Scan-report contrastive loss
- Localization loss
- Prompt-based disease label loss



RefCT dataset, Uniklinik Freiburg

12

years

98k

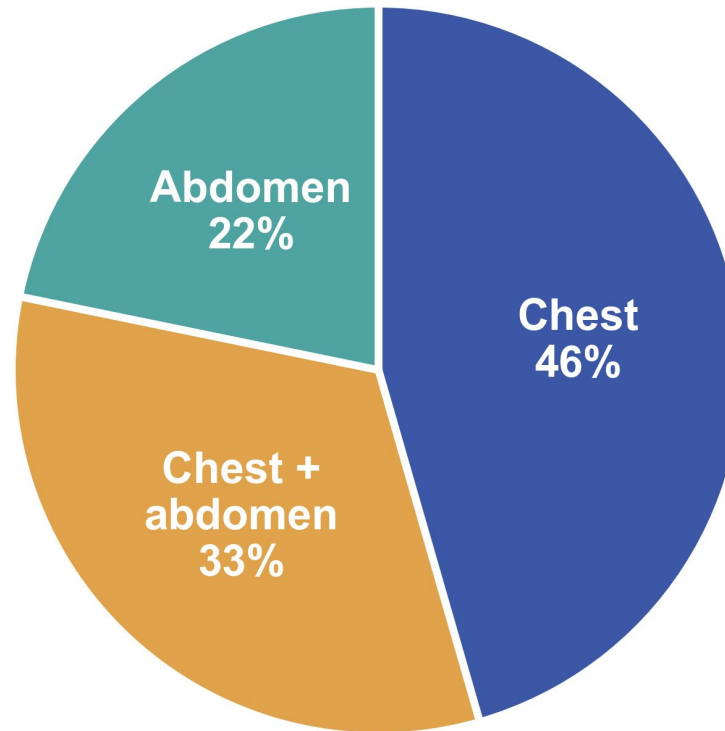
report-scan pairs

50k

patients

262k

snippet-slice pairs



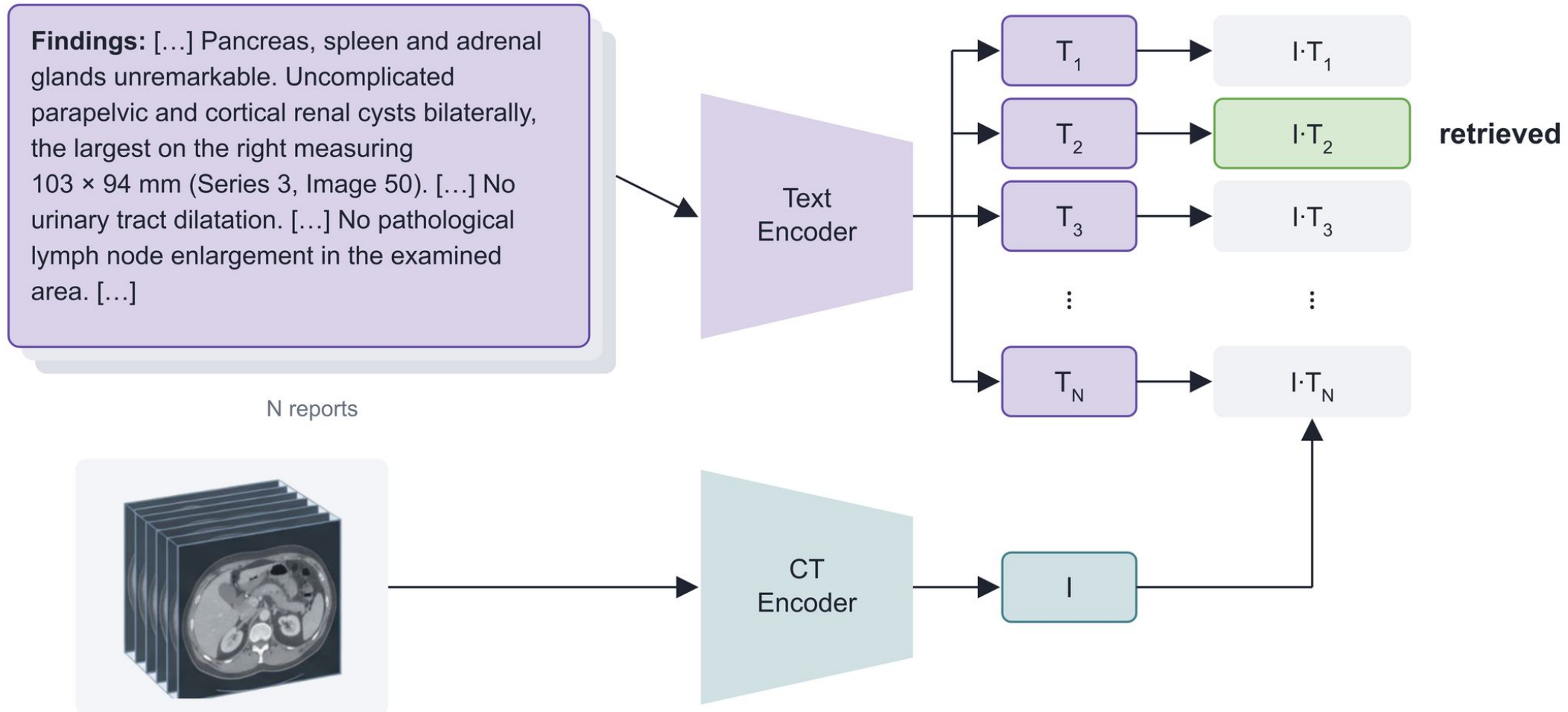
Training setup

- Combine RefCT with chest/abdomen public datasets
- Train for 32 hours on 1xH100 GPU

Training set	Report-scan pairs
RefCT (ours)	78k
CT-RATE	24k
Merlin	15k
Inspect	19k
Total	136k

Inference task: Retrieval

- Given a scan, **retrieve** similar scans or reports



Results: Retrieval

Method	CT-RATE		Merlin	
	R@10 $\uparrow$		R@1 $_{\text{find}}$ $\uparrow$	
CT-CLIP	5.0		—	
MedVista3D-ViT	10.7		—	
Merlin	2.7		59.4	
BrgSA	22.2		—	
MPS-CT	16.3		—	
SPECTRE	18.2	1.4	44.5	0.7
RadFinder (Ours)	31.5	1.6	69.0	0.9

Inference task: Text-based disease classification

Simple renal cysts

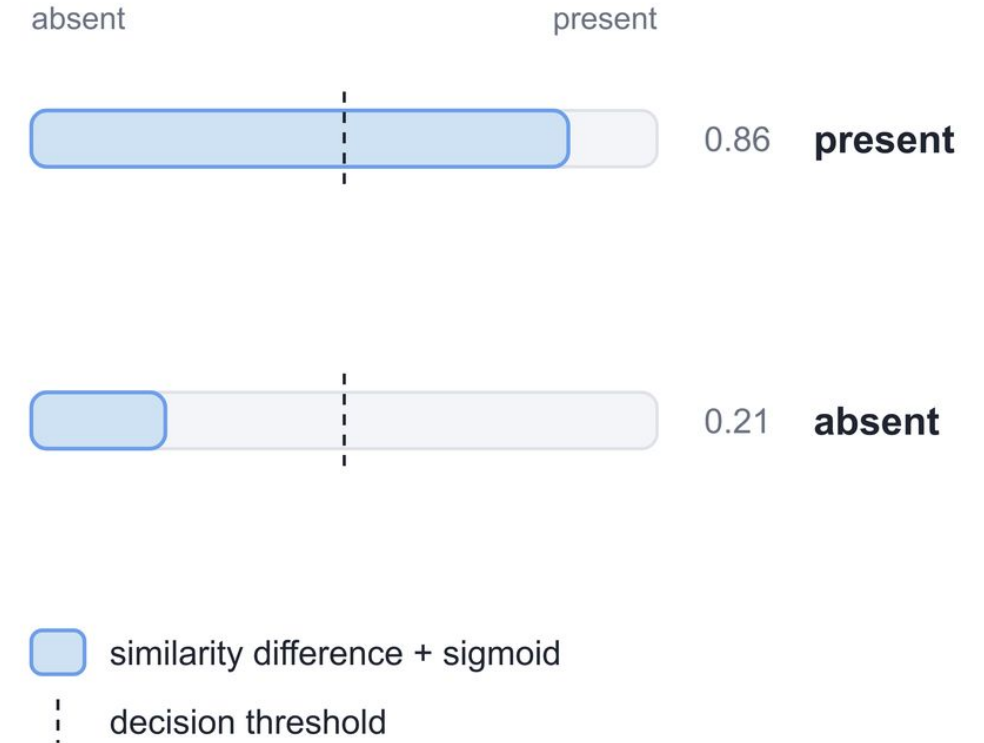
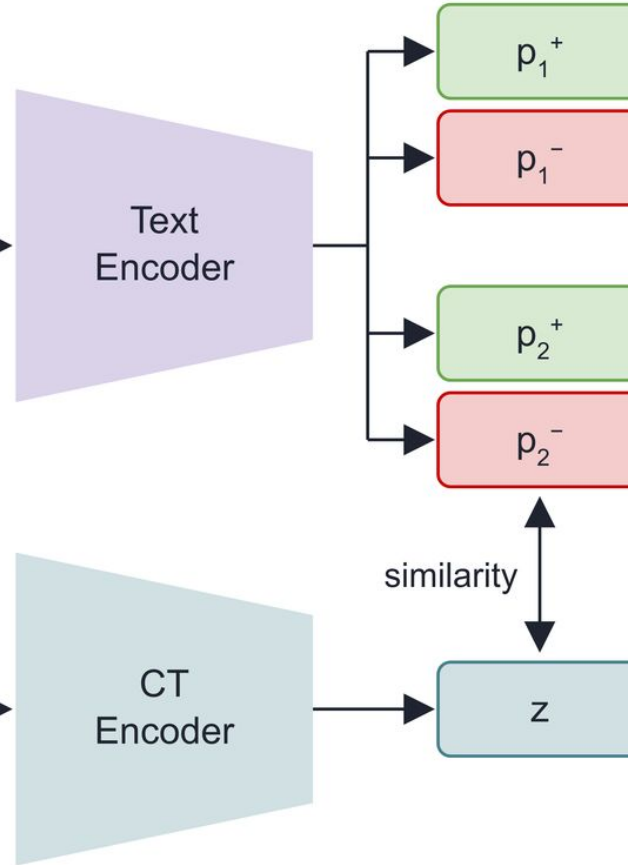
Simple renal cysts are present.

There are no simple renal cysts.

Hydronephrosis

Hydronephrosis is present.

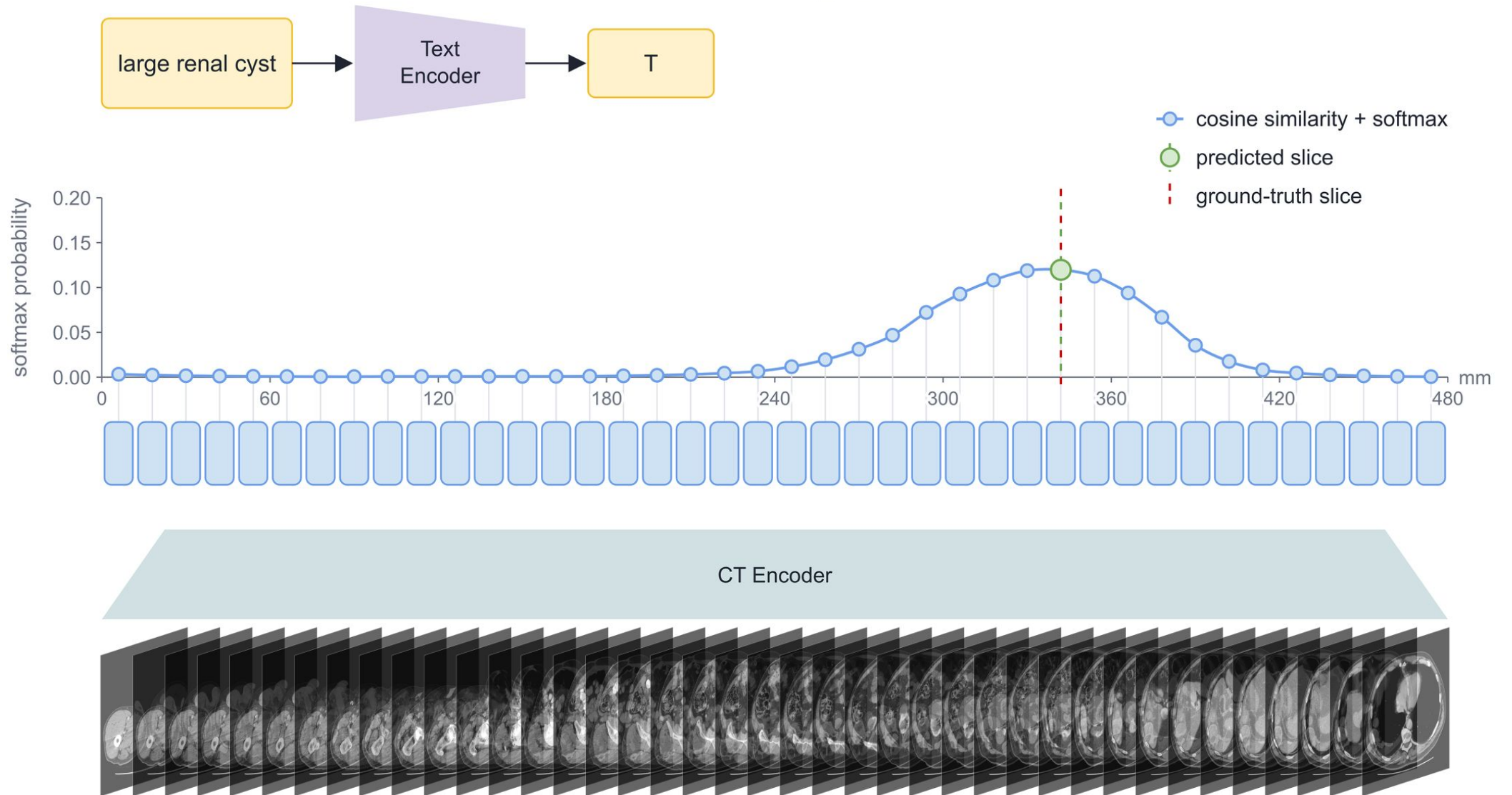
There is no hydronephrosis.



Results: Disease classification

Method	CT-RATE		RC	
	AUC↑		AUC↑	
CT-CLIP	73.1		62.9	
MedVista3D-ViT	77.8		—	
Merlin	72.8		64.4	
BrgSA	82.9		74.2	
MPS-CT	83.8		77.3	
SPECTRE	54.3	1.1	55.7	0.6
RadFinder (Ours)	83.8	0.7	77.0	0.5

Inference task: Localization



Results: Localization

Method	RefCT MAE↓	
Random slice	126.9	5.7
Middle slice	95.8	3.6
CLIP ViT-B/32	124.6	5.5
BiomedCLIP	86.6	5.2
MedSigLIP-448	75.6	5.0
SigLIP2	82.4	4.7
SigLIP2 finetuned	67.0	4.9
RadFinder (Ours)	36.3	2.9

Ablation: Localization loss

Method	CT-RATE						RC		Merlin		RefCT	
	R@10↑		MAP@5↑		AUC↑		AUC↑		R@1 _{find} ↑		MAE↓	
All losses	31.5	1.6	78.0	1.7	83.8	0.7	77.0	0.5	69.0	0.9	36.3	2.9
No localization loss	31.5	1.6	78.2	1.7	83.6	0.7	76.9	0.5	69.3	0.9	128.8	5.6

- Localization loss does not influence the other tasks

Ablation: Prompt-based disease label loss

Method	CT-RATE				RC				Merlin	
	R@10↑		MAP@5↑		AUC↑		AUC↑		R@1 _{find} ↑	
All losses	31.5	1.6	78.0	1.7	83.8	0.7	77.0	0.5	69.0	0.9
Global loss only	29.4	1.6	76.6	1.7	56.9	1.1	62.8	0.6	70.7	0.7

- Prompt loss enables disease classification
- Retrieval mostly unaffected

Ablation: Data mixture

Method	CT-RATE						RC		Merlin	
	R@10↑		MAP@5↑		AUC↑		AUC↑		R@1 _{find} ↑	
All datasets	31.5	1.6	78.0	1.7	83.8	0.7	77.0	0.5	69.0	0.9
RefCT dataset only	26.3	1.6	76.7	1.7	80.4	0.8	77.7	0.6	60.5	0.9
Public datasets only	26.9	1.6	77.1	1.7	83.2	0.7	73.6	0.6	67.5	0.9

- Internal data alone transfers well to public datasets
- Our training setup also works on public datasets
- Both contributions are complementary

Conclusion

- Radiology reports contain various supervision: scan description, disease classes, and locations of findings
- Our RadFinder model learns all three tasks at once without degrading on either task

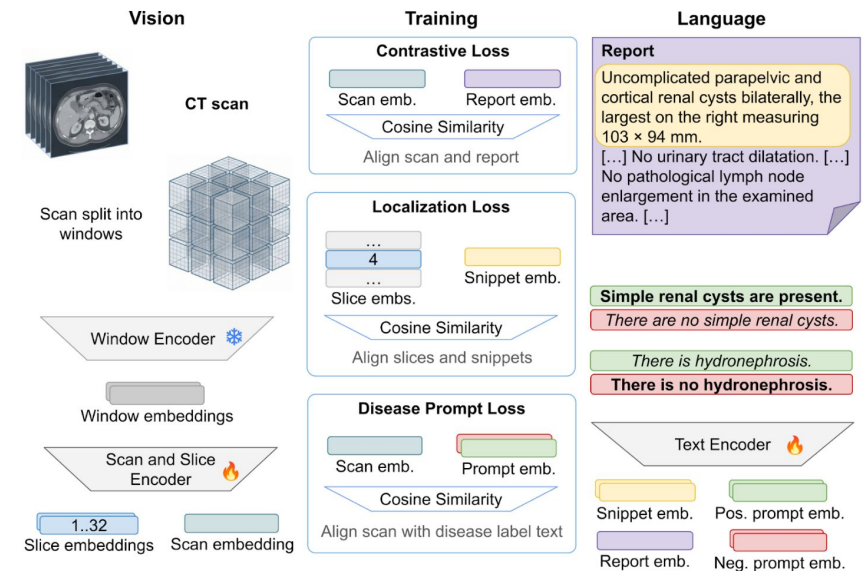
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report

Simple renal cysts: **Yes**

Hydronephrosis: **No**

Snippet



Thank you!

Code and models:



radfinder.github.io



Simon Ging



Philipp Arnold



Sebastian Walter



Hani Alnahas



Hannah Bast



Elmar Kotter



Jiancheng Yang



Behzad Bozorgtabar



Thomas Brox



Backup slides

Training setup details

- Our training set and chest/abdomen public datasets
- 32 hours on a single H100
- Batch size 4k, 10 Epochs, AdamW, LR $2e-4$, 1 Epoch warmup, cosine decay
- Initi. from SPECTRE.
 - ViT-L local backbone, frozen
 - 4-layer global combiner
 - Qwen3-Embedding 0.6B text encoder with LoRA
 - SigLIP heads into a 512-d shared space
 - Full volumes with variable input shapes (SPECTRE crops to a fixed grid)
- Text augmentation: 20% strip prior exams, 20% drop findings section, 20% replace with concatenated organ-level descriptions from RATE.

Training set	Report-scan pairs
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Total	136k

Full retrieval and disease classification results

	Data				CT-RATE				RC		Merlin R@1					
	A	B	C	D	R@10		MAP@5	AUC		AUC		Findings		Impr.		
Random Chance					0.3		0.1	50.0		50.0		0.8		0.8		
CT-CLIP [6]	✓				5.0		68.3	73.1		62.9		—		—		
MedVista3D-ViT [11]	✓				10.7		—	77.8		—		—		—		
MedVista3D-UniMISS	✓				8.7		—	78.2		—		—		—		
Merlin [2]			✓		2.7		62.6	72.8		64.4		59.4		19.4		
fVLM* [16]	✓				—		—	77.8		68.0		—		—		
BrgSA [9]	✓				22.2		70.4	82.9		74.2		—		—		
MPS-CT [10]	✓				16.3		71.3	83.8		77.3		—		—		
SPECTRE [3]	✓	✓	✓		18.2	1.4	76.8	1.7	54.3	1.1	55.7	0.6	44.5	0.7	29.2	0.6
RadFinder (Ours)	✓	✓	✓	✓	31.5	1.6	78.0	1.7	83.8	0.7	77.0	0.5	69.0	0.9	40.3	0.7
No localization loss	✓	✓	✓	✓	31.5	1.6	78.2	1.7	83.6	0.7	76.9	0.5	69.3	0.9	40.2	0.6
Global loss only	✓	✓	✓	✓	29.4	1.6	76.6	1.7	56.9	1.1	62.8	0.6	70.7	0.7	42.2	0.7
Prompt loss only	✓	✓	✓	✓	5.6	0.8	77.9	1.7	84.8	0.7	78.6	0.5	13.1	0.6	10.9	0.4
RefCT dataset	✓				26.3	1.6	76.7	1.7	80.4	0.8	77.7	0.6	60.5	0.9	35.8	0.7
Public datasets, no loc.	✓	✓	✓		26.9	1.6	77.1	1.7	83.2	0.7	73.6	0.6	67.5	0.9	40.2	0.8

Training datasets: A: RefCT (ours, 78k), B: CT-RATE [6] (47k), C: Merlin [2] (15k), D: INSPECT [7] (19k).

*fVLM additionally trains on MedVL-CT69 [16] (272k).

Full localization results

	MAE (mm)		<6mm		<18mm		<30mm	
Random slice	126.9	5.7	4.7	1.1	11.8	1.6	17.8	2.0
Middle slice	95.8	3.6	4.6	1.2	13.0	1.8	20.8	2.2
CLIP ViT-B/32 [13]	124.6	5.5	3.2	0.8	9.4	1.4	14.8	1.7
BiomedCLIP [21]	86.6	5.2	8.3	1.3	20.2	1.9	30.5	2.2
MedSigLIP-448 [15]	75.6	5.0	9.7	1.4	26.2	2.1	40.1	2.3
SigLIP2 [17]	82.4	4.7	7.8	1.3	19.3	1.8	31.5	2.3
SigLIP2 finetuned*	67.0	4.9	17.4	1.8	35.8	2.4	48.7	2.3
RadFinder (Ours)	36.3	2.9	20.3	2.0	45.3	2.5	61.8	2.4
Loc. loss only	36.0	2.9	19.9	2.0	45.2	2.5	62.7	2.4

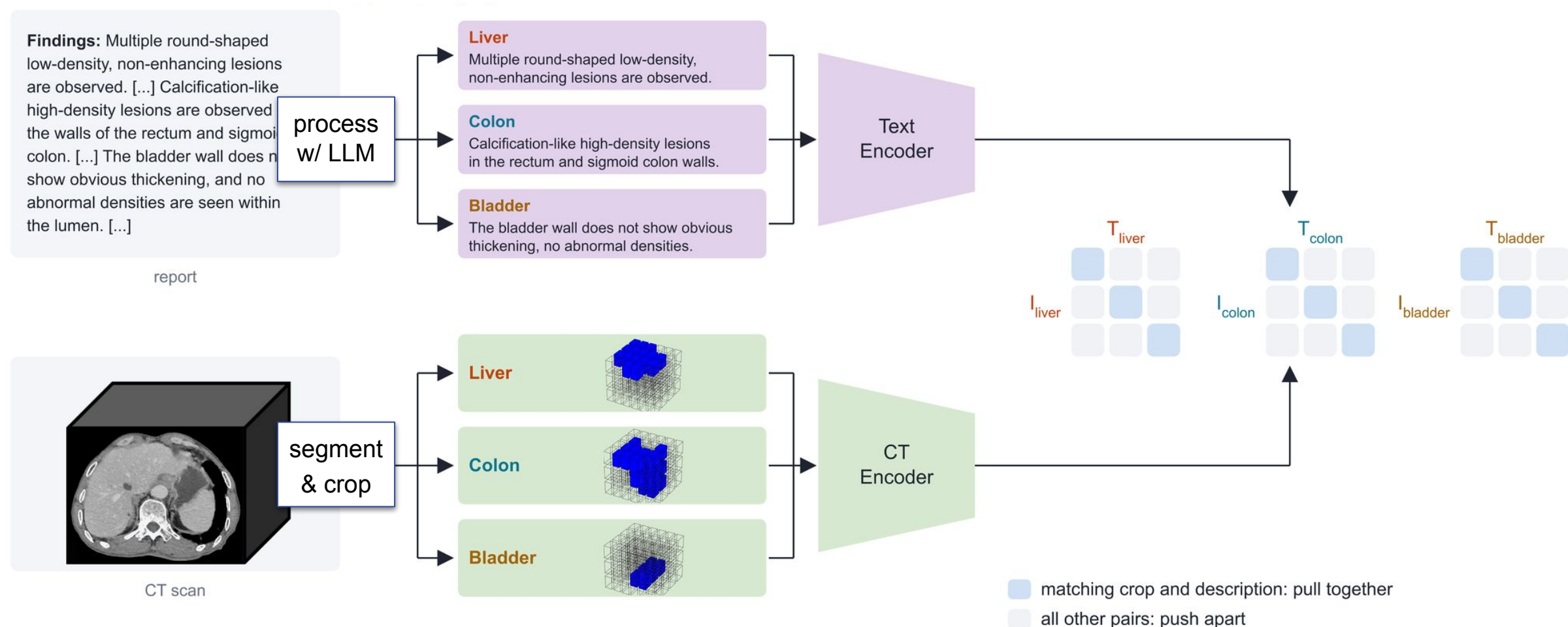
Table 2. Results for snippet localization on the RefCT test set. *SigLIP2 finetuned on RefCT snippet–slice pairs via contrastive loss.

Full ablations table

Method	CT-RATE				RC				Merlin	
	R@10		MAP@5		AUC		AUC		R@1 _{find}	
RadFinder (Ours)	31.5	1.6	78.0	1.7	83.8	0.7	77.0	0.5	69.0	0.9
No localization loss	31.5	1.6	78.2	1.7	83.6	0.7	76.9	0.5	69.3	0.9
Global loss only	29.4	1.6	76.6	1.7	56.9	1.1	62.8	0.6	70.7	0.7
Prompt loss only	5.6	0.8	77.9	1.7	84.8	0.7	78.6	0.5	13.1	0.6
RefCT dataset only	26.3	1.6	76.7	1.7	80.4	0.8	77.7	0.6	60.5	0.9
Public datasets only	26.9	1.6	77.1	1.7	83.2	0.7	73.6	0.6	67.5	0.9

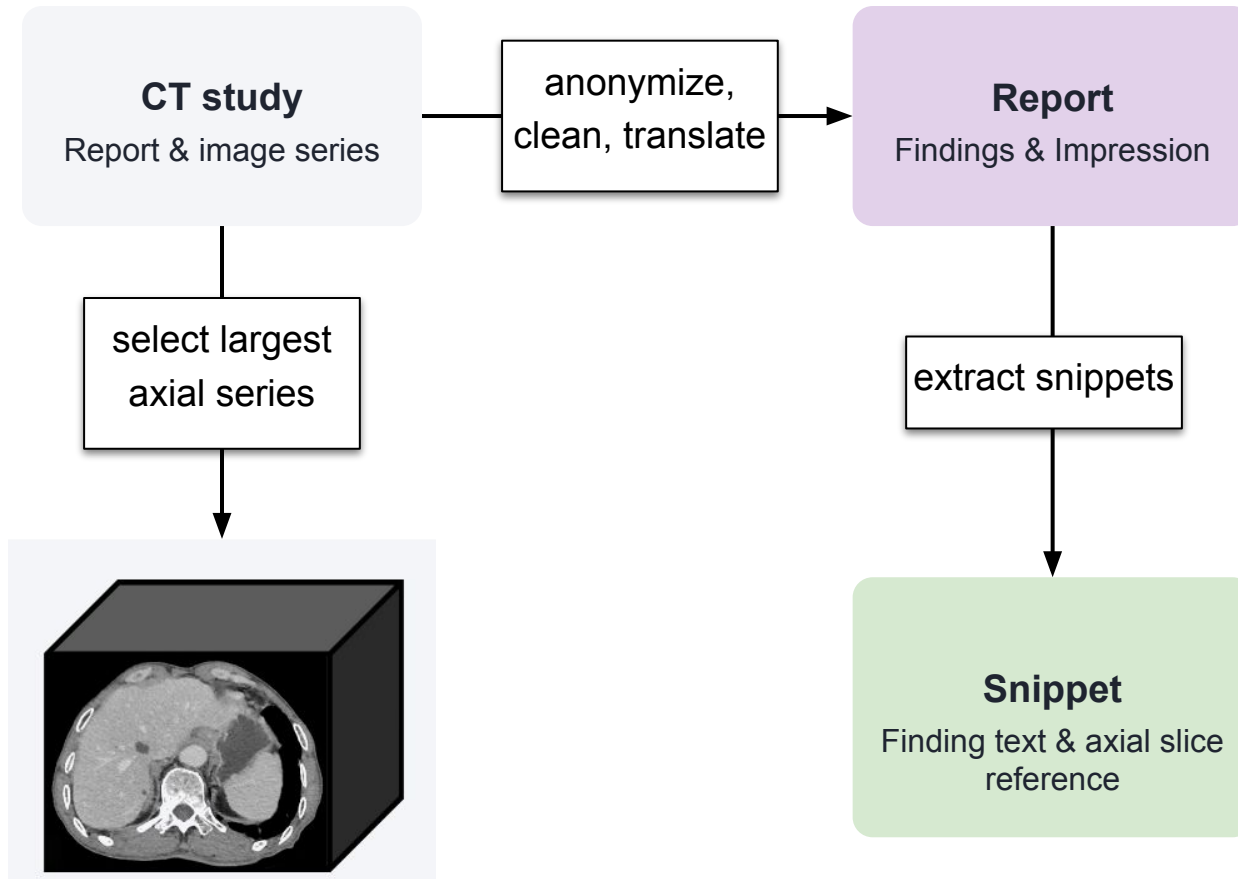
Radiology VLMs: organ supervision

- Other models align segmented organs and extracted organ texts



Shui et al., Large-scale and Fine-grained Vision-language Pre-training for Enhanced CT Image Understanding, ICLR 2025

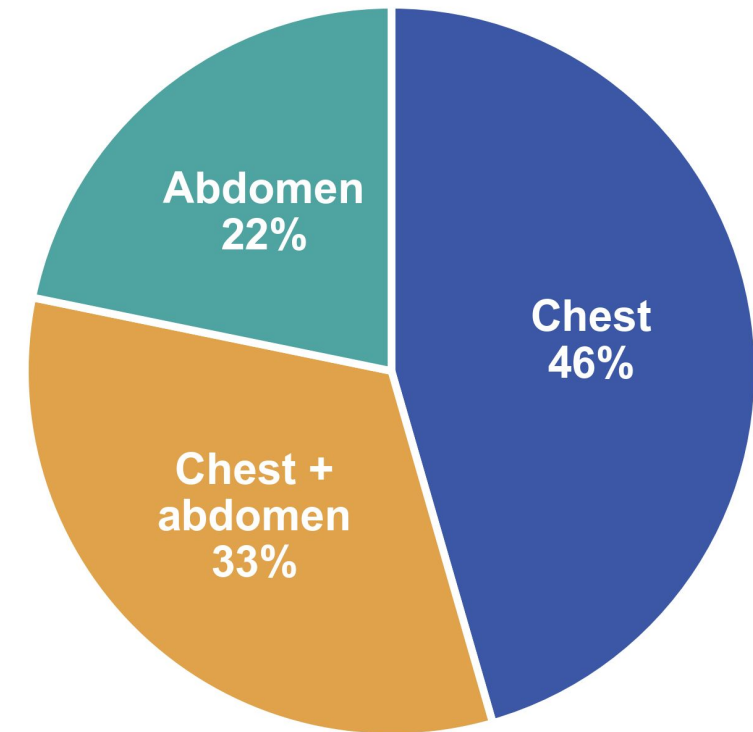
RefCT dataset, Uniklinik Freiburg, chest & abdomen



98k
report–volume pairs

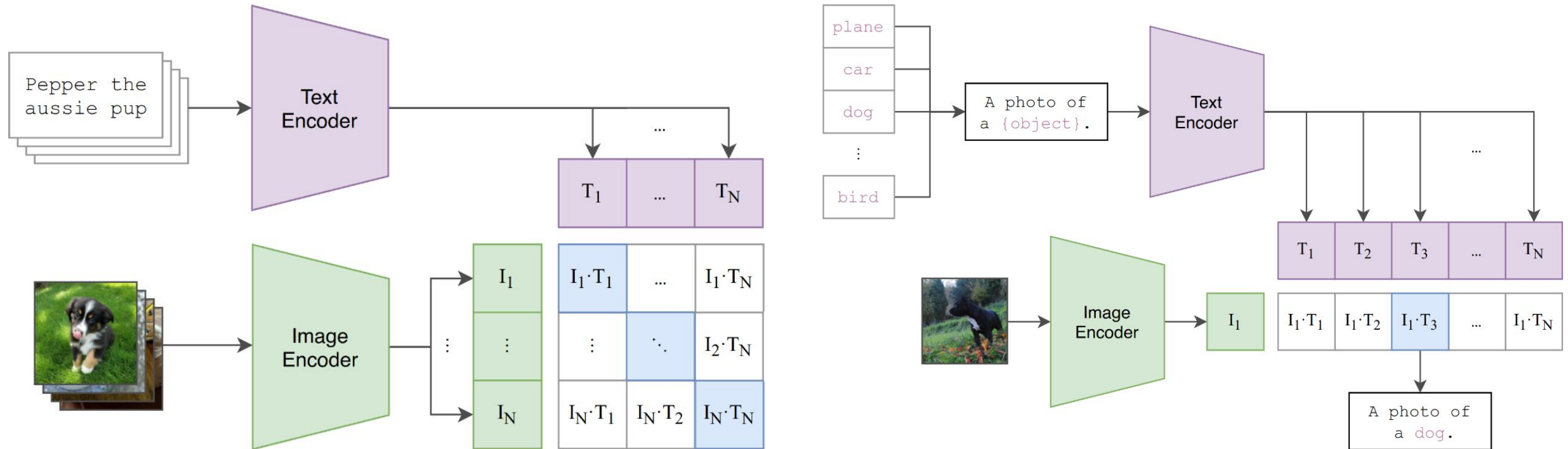
50k
patients

262k
snippet–slice pairs



Contrastive learning is a popular approach to learn features

- CLIP models learn embeddings by contrasting images and text
- These embeddings can then be used for, e.g., zero-shot classification



Radford et al. (2021) Learning transferable visual models from natural language supervision. ICML 2021, 139:8748-8763

